Rotary Wing Aerodynamics (AE 667)

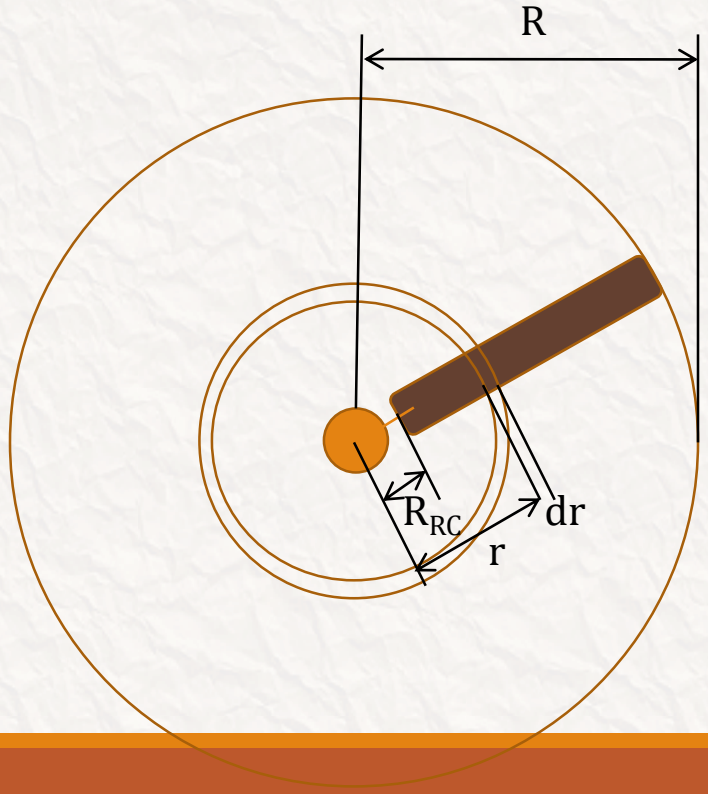


Blade Element (BE) Theory

Blade Element Theory

- ‘Strip’ theory
  - Divide blade into thin strips
  - Compute lift, drag, etc. at the strips
  - Integrate values at the strips to get the total thrust, torque, power
- Assumes uniform induced velocity (v)

$$T = b\int_{R_{RC}}^{R} dT\,; \quad P = b\int_{R_{RC}}^{R} dP$$



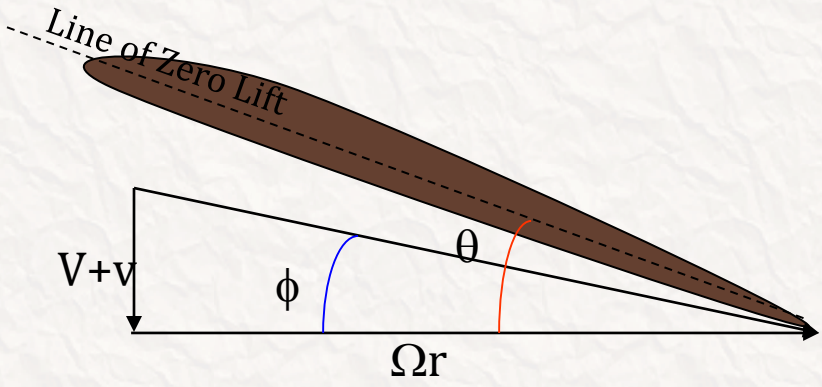
Blade section view

$\theta = \text{Local blade pitch angle}$

$\alpha = \text{Angle of attack}$

$$\phi = \tan^{-1}\left(\frac{V + \mathrm{v}}{\Omega r}\right)$$

$$\alpha_{effective} = \theta - \phi = \theta - \tan^{-1}\left(\frac{V + \mathrm{v}}{\Omega r}\right)$$

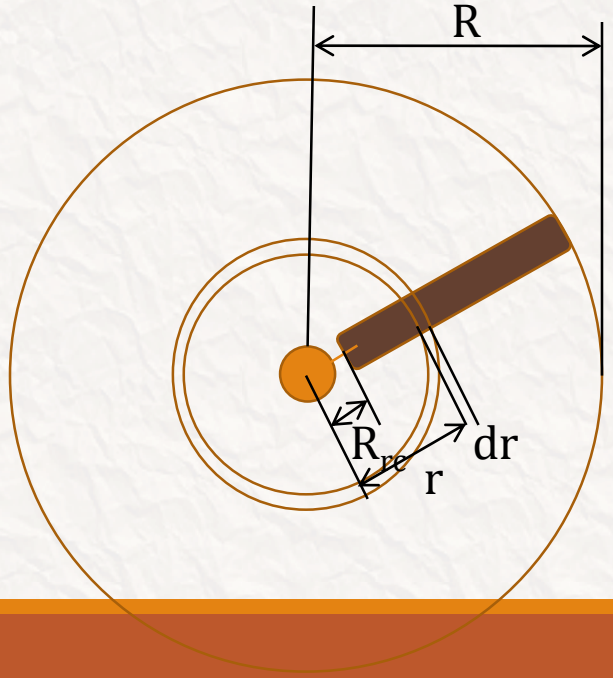


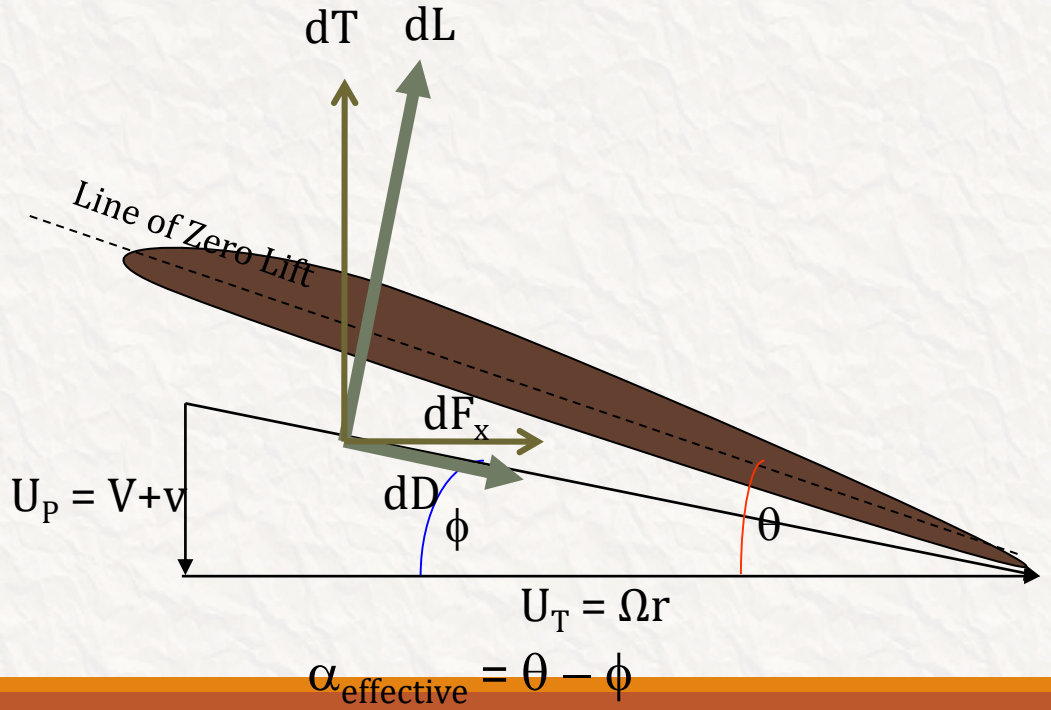
$\alpha_{effective} = \theta - \phi$

Blade section forces

Nomenclature

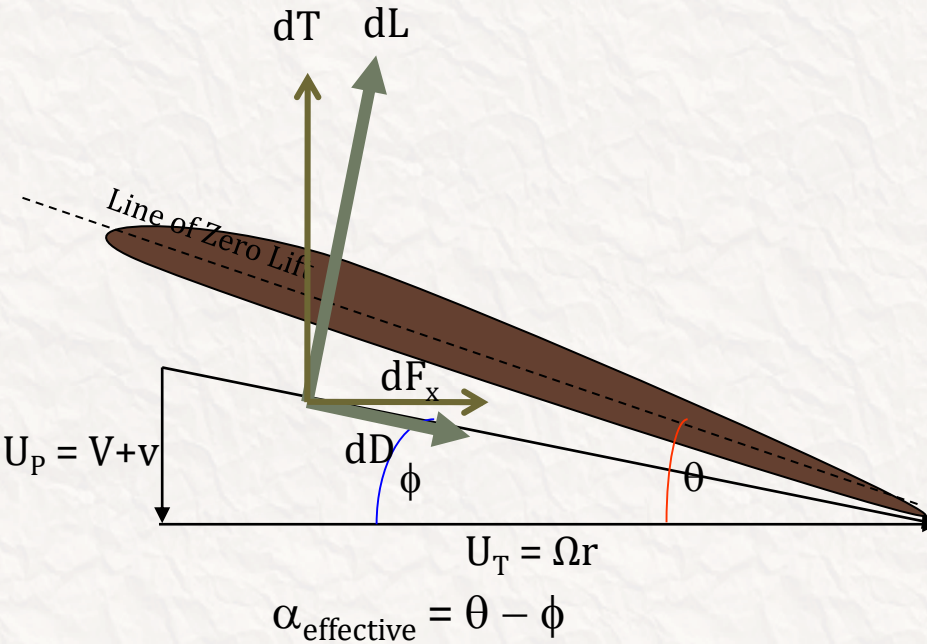
- dL: Sectional airfoil lift
- dD: Sectional airfoil Drag
- dT: Sectional thrust
- dF_x: Force in X direction (tangential to rotor disk)
- U_T: Tangential Velocity = Ωr
- U_P: Perpendicular Velocity = V+v





Blade section forces

- U_T: Tangential Velocity = Ωr
- U_P: Perpendicular Velocity = V+v



$$dL = \frac{1}{2}\rho(U_T^2 + U_P^2)\, cC_l$$

$$dD = \frac{1}{2}\rho(U_T^2 + U_P^2)\, cC_d$$

$$dT = (dL\cos(\phi) - dD\sin(\phi))dr$$
$$= \frac{1}{2}\rho(U_T^2 + U_P^2)c(C_l\cos(\phi) - C_d\sin(\phi))dr$$

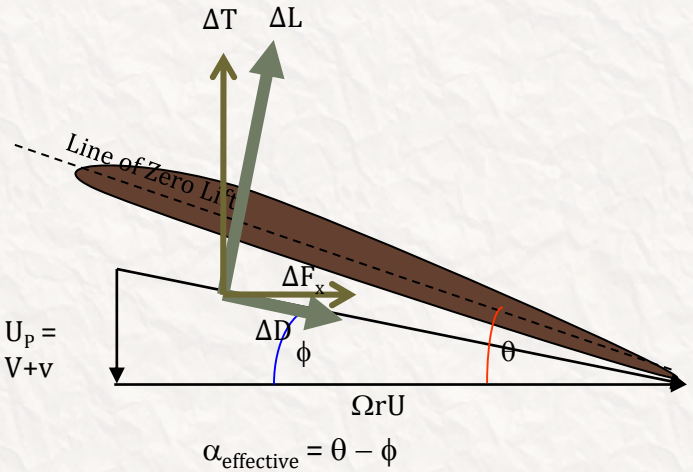
$$dF_x = (dD\cos(\phi) + dL\sin(\phi))dr$$
$$= \frac{1}{2}\rho(U_T^2 + U_P^2)c(C_d\cos(\phi) + C_l\sin(\phi))dr$$

$$dP = U_T dF_x = \Omega r dF_X$$

Rotor Forces

$$T = b \int_{R_{RC}}^{R} dT \;;$$

$$= b \int_{R_{RC}}^{R} \frac{1}{2}\rho(U_T^2 + U_P^2)c(C_l \cos(\phi) - C_d \sin(\phi))dr$$



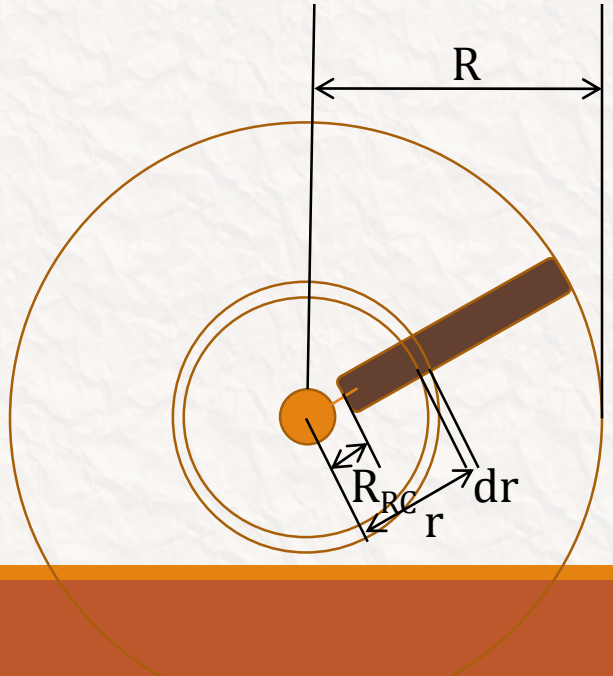
$$Q = b \int_{R_{RC}}^{R} r dF_x \quad \text{(Torque)}$$

$$= b \int_{R_{RC}}^{R} r\frac{1}{2}\rho(U_T^2 + U_P^2)c(C_d \cos(\phi) + C_l \sin(\phi))dr$$

$$P = b \int_{R_{RC}}^{R} \Omega r dF_X$$

$$= b \int_{R_{RC}}^{R} \Omega r\frac{1}{2}\rho(U_T^2 + U_P^2)c(C_d \cos(\phi) + C_l \sin(\phi))dr$$

$$= Q\Omega$$



For analytical integration

Assumptions:

- Constant chord (c)
- $(V+v) << \Omega r \therefore \Phi << 1, \cos(\Phi) \approx 1, \ \sin(\Phi) \approx \Phi$
- Linearly varying Coefficient of lift

$$(C_l) = a(\alpha_{eff}) = a(\theta - \Phi)$$

- $C_d << 1$, therefore $C_d \sin(\Phi) \approx 0$

$$T = b \int_{R_{RC}}^{R} \frac{1}{2}\rho(U_T^2 + U_P^2)c(C_l \cos(\phi) - C_d \sin(\phi))dr \approx \frac{1}{2}\rho cba\Omega^2 \int_{r=0}^{r=R} \left(\theta - \frac{V}{\Omega r} - \frac{v}{\Omega r}\right) r^2 dr$$

$$P = b \int_{R_{RC}}^{R} \Omega r \frac{1}{2}\rho(U_T^2 + U_P^2)c(C_d \cos(\phi) + C_l \sin(\phi))dr$$

$$P \approx \frac{1}{2}\rho cba\Omega^3 \int_{r=0}^{r=R} \left[\left(\theta - \frac{V}{\Omega r} - \frac{v}{\Omega r}\right)\left(\frac{V}{\Omega r} + \frac{v}{\Omega r}\right) + C_d\right] r^3 dr$$

For analytical integration

- More assumptions:
  - Linear blade twist: θ = E + Fr

$$T = \frac{b}{2}\rho\Omega^2 ca\left[\frac{1}{3}\left(E + \frac{3}{4}FR\right) - \frac{V+v}{2\Omega R}\right]R^3 = \frac{b}{2}\rho ca(\Omega R)^2 R\left[\frac{\theta_{.75R}}{3} - \lambda/2\right]$$

$$C_T = \frac{abc}{2\pi R}\left[\frac{\theta_{.75}}{3} - \lambda/2\right] = \frac{a\sigma}{2}\left[\frac{\theta_{.75R}}{3} - \lambda/2\right] = \frac{a\sigma}{4}\left[\frac{4\theta_{.75R}}{3} - \lambda\right]$$

where

$$\lambda = \text{Inflow Ratio} = \frac{V+v}{\Omega R}$$

Notice that C_T is linearly proportional to pitch at 75% radius.
Hence, pitch angle is generally defined at r = 0.75R

For analytical integration

- Power can be integrated in the similar fashion.
  - If C_d is assumed to be constant along rotor,

$$C_P = \lambda C_T + \frac{\sigma C_d}{8}$$

FM for BET rotor

$$FM = \frac{\text{Ideal power through Momentum Theory}}{\text{Power through Blade Element Theory}}$$

$$= \frac{\lambda C_T}{\lambda C_T + \sigma C_d/8}$$

- High solidity (wide blades, more blades) ➔ low FM
- High C_d (high airfoil drag) ➔ low FM
- High C_T ➔ high FM

(caution!) high FM ≠ better efficiency

Computing average C_l of rotor blade
(Equivalent constant $\bar{C}_l$ across the blade)

Assuming untapered (c = constant)

$$T = b\int_0^R \frac{1}{2}\rho c(\Omega r)^2\,\bar{C}_l\,dr = \frac{\rho bc\bar{C}_l\Omega^2R^3}{6}\ ;$$

$$C_T = \frac{T}{\rho\pi R^2(\Omega R)^2} = \frac{bc}{\pi R}\frac{\bar{C}_l}{6} = \sigma\frac{\bar{C}_l}{6}$$

$$\bar{C}_l = 6\frac{C_T}{\sigma} \quad \text{where } \bar{C}_l = \text{Average Lift Coefficient}$$

Defining a few more terms…

- Average/mean Lift Coefficient = $\bar{C}_l = 6\frac{C_T}{\sigma}$

- $\frac{C_T}{\sigma} = \frac{Thrust}{\rho(Blade\ area)(\Omega R)^2}$ ➔ **Blade Loading**

- $\frac{\bar{C}_l}{a} = \frac{6C_T}{\sigma a}$ ➔ **Mean Angle of Attack**

Limitations

- Does not account for swirl losses
- Assumes uniform induced velocity
- Does not account for tip losses
  - Tip loss factor (as a function of radius) can be included in thrust expression while integrating

Empirical expression for power :

$$C_P = \kappa\lambda C_T + \sigma\frac{C_d}{8} \quad \text{where } k \approx 1.15$$